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CCH Tagetik Installation Manual for WildFly and JBoss EAP

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Introduction

This manual covers the procedures for installing and configuring the CCH Tagetik application on WildFly or JBoss EAP as a single node.
Please check the “Technical Architecture: CCH Tagetik Compatibility Matrix – Server” document for compatibility information about the service pack of CCH Tagetik which you would like to deploy.

The intended audience of this manual is anyone who needs to understand and install CCH Tagetik on the WildFly or JBoss EAP compliant application server.
This manual contains the necessary steps to complete a basic deployment of the application without going into details of the specific AS version and flavor. For this reason, it is not intended to substitute the documentation provided by the developers and vendors of the specific AS.

The deployment of the application requires valid data sources, and this manual assumes that the valid data sources are available. Instructions on how to create these data sources are provided in a specific manual available on the CCH Tagetik community website.

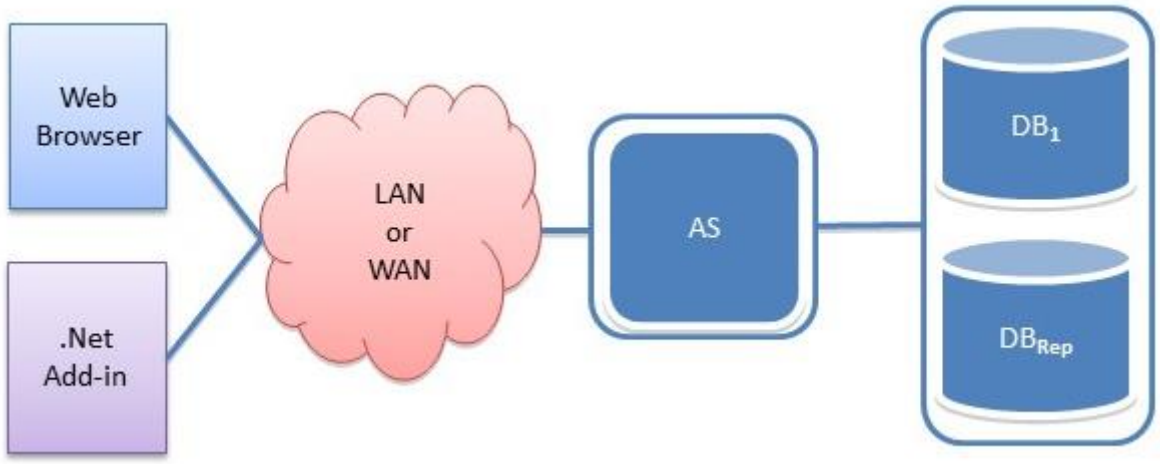
Please contact CCH Tagetik concerning any questions or comments related to the contents of this document.

Overview

The installation of the application on the server tiers consists of two primary steps:

1. the creation of the application databases/schemas on a compatible RDBMS and,
2. the configuration of the JEE application server followed by the deployment of the application EAR archive onto a JEE compliant Application Server.

This installation manual focuses on the installation procedure for step 2 and assumes that the application databases have already been created before testing the application deployment.



The basic architecture of the application.

The basic architecture of the application is shown in Figure 1. This manual covers the application server (AS) tier in the above figure. Step 1 of the installation procedure relates to the database server tier (DB). Step 2 can be further divided into the following sub-steps:

1. Check the prerequisites before the installation procedure.
2. Configure the data sources which allow the AS to communicate with the RDBMS.
3. Configure, if necessary, the ports to be used by the AS interfaces and for the application.
4. Configure WildFly/JBoss EAP and the application server logging.
5. Deploy the application EAR file on the AS

Prerequisites

To install, configure, and deploy the application the following prerequisites must be satisfied.
The following checklist is provided for the reader.

- ✓ **CCH Tagetik Application Version:** Please check the compatibility matrices for compatibility with WildFly or JBoss EAP.
- ✓ **WildFly or JBoss EAP**: A compatible version of WildFly or JBoss EAP should be installed. Please check the CCH Tagetik compatibility matrices for compatibility with the WildFly or JBoss EAP
- ✓ **AS-DB Communication Open**: the AS host server should be able to communicate to the RDBMS host server. If there is a firewall between the two servers, the appropriate ports should be opened, and traffic permitted depending on the configuration of the RDBMS.
- ✓ **JDK Installed:** A compatible version of a JDK must be installed on the application server by a user with administrative privileges. Please check the "Technical Architecture: CCH Tagetik Compatibility Matrix – Server" document for compatibility with WildFly or JBoss EAP.
  NOTE: when using JDK 11 on Linux the following OS packages must be installed: fontconfig, libfontconfig1, libasound2, libasound2-data.
- ✓ **CCH Tagetik EAR File**
- ✓ **DBs Installed**: CCH Tagetik database/schema should be created with the necessary access configurations on a compatible RDBMS. This part of the installation procedure is covered in a separate manual.
- ✓ **Adequate AS Host Resources:** Please refer to the "Technical Architecture: CCH Tagetik 5.3 Hardware & Software Architecture Overview" document. In general, customer projects require more than the minimum to provide adequate user response times. If you have not yet received an estimate of the recommended resources for the project, please contact CCH Tagetik or the project manager to request one.

Configuration

For this step you will need WildFly or JBoss EAP installed according to the documentation of the AS currently being used.
The official WildFly documentation can be found at https://docs.wildfly.org/ while the JBoss EAP one can be found at https://access.redhat.com/documentation/en-us/red_hat_jboss_enterprise_application_platform.

Step 1 – Install JDBC Drivers on WildFly/JBoss EAP

To configure databases, you need to configure drivers. Download the JDBC driver from your database vendor. Please see the "Technical Architecture: CCH Tagetik Compatibility Matrix – Server" document for drivers compatibility.
The following is an example of a manual installation, but you can install JDBC drivers also through:

1. WildFly management console
2. JBoss CLI console

For these alternative methods please refer to the WildFly or JBoss EAP official documentation (e.g. WildFly documentation for DataSource configuration can be found at https://docs.wildfly.org/26.1/Admin_Guide.html#DataSource, please check the documentation version in the URL).

Manual Procedure

1. Create a *"tgk"* folder under the *<JBOSS_HOME>or<WILDFLY_HOME>\modules* directory.
2. Create a file path structure under the *<JBOSS_HOME>or<WILDFLY_HOME>\modules\tgk* directory. For instance, for a SQL SERVER driver, create the following directory structure: *<JBOSS_HOME> or<WILDFLY_HOME>\modules\tgk\com\microsoft\sqlserver\jdbc\main.*
3. Copy the JDBC driver JAR file into the *main* subdirectory.
4. In the main subdirectory, create a *module.xml* file.
   Example for MS SQL SERVER driver:

```
<module xmlns="urn:jboss:module:1.3" name="tgk.com.microsoft.sqlserver.jdbc">
    <resources>
        <resource-root path="mssql-jdbc-x.x.x.jar"/>
    </resources>
    <dependencies>
        <module name="javax.api"/>
        <module name="javax.transaction.api"/>
    </dependencies>
</module>
```

Note:

The module XSD and JDBC driver versions defined above must be intended as an example. The actual module XSD version is defined in the <JBOSS_HOME>or<WILDFLY_HOME>\docs\schema\module-X.xsd file.

5. Add the driver in the standalone configuration:
   Edit the *<JBOSS_HOME>or<WILDFLY_HOME>\standalone\configuration\standalone.xml* file adding a new <driver> element under the <drivers> tag.
   Example for MS SQL SERVER driver:

```
<drivers>
    <driver name="sqlserver" module="tgk.com.microsoft.sqlserver.jdbc">
        <driver-class>com.microsoft.sqlserver.jdbc.SQLServerDriver</driver-class>
    </driver>
</drivers>
```

Repeat this procedure for any other database vendor to be used.

Step 2 – Data sources configuration.

In the previous step JDBC drivers have been installed, now the application server needs the appropriate references for the database connections.

For this step you will need to launch the RDBMS and to have a Tagetik database already created. Please check the specific manual to create one or more databases.

The following is an example of a manual installation, but you can configure datasources also through:

1. WildFly management console
2. JBoss CLI console

For these alternative methods please refer to the WildFly or JBoss EAP official documentation (WildFly documentation for DataSource configuration can be found at https://docs.wildfly.org/26.1/Admin_Guide.html#DataSource, please check the documentation version in the URL).

Manual Procedure

Go to *<JBOSS_HOME>or<WILDFLY_HOME>\standalone\configuration* and edit file *standalone.xml*.
Under the <datasources> tag, for each datasource, add a <datasource> tag with a specific *jndi-name* as follows:

- For the REPOSITORY database use:
  "java:/DBEASYFINANCE2_REPOSITORY"
- For applicative data sources use:
  "java:/DBEASYFINANCE2_D00[0-23] " format (e.g. "java :/DBEASYFINANCE2_D0000 ", "java :/DBEASYFINANCE2_D0001", etc.)
- For secondary data sources use:
  "java:/DBEASYFINANCE2_SECONDARY_00[0-25]" format (e.g. "java:/DBEASYFINANCE2_SECONDARY_0000", "java:/DBEASYFINANCE2_SECONDARY_0001", etc.)

Example for MS SQL SERVER REPOSITORY datasource with Wildfly for JAVAEE:

```
<datasource jndi-name="java:/DBEASYFINANCE2_REPOSITORY"
        pool-name="java:/DBEASYFINANCE2_REPOSITORY">
    <connection-url>jdbc:sqlserver://db-test3:1433;databaseName=DATABASE_NAME;
;encrypt=true;trustServerCertificate=true</connection-url>
    <driver>sqlserver</driver>
    <pool>
        <min-pool-size>0</min-pool-size>
        <max-pool-size>100</max-pool-size>
    </pool>
    <security>
        <user-name>USERNAME</user-name>
```

```
        <password>PASSWORD</password>
    </security>
    <validation>
        <check-valid-connection-sql>select ''</check-valid-connection-sql>
    </validation>
    <timeout>
        <idle-timeout-minutes>5</idle-timeout-minutes>
    </timeout>
</datasource>
```

Attention: if you are using a WIldfly/JBoss JakartaEE version the security tag is inline:

```
<security user-name="tagetikcpm" password="tagetikcpm"/>
```

Make sure that the name of the driver inside the <driver> tag is equal to the name of the actual driver.

Note: WildFly requires that all the resources defined in the EAR correspond to a specific JEE/JakartaEE data resource on the Application Server. For each datasource not mapped to a real database please add the following snippet:

```
<datasource jndi-name="java:/DBEASYFINANCE2_D000X" pool-name="java:/DBEASYFINANCE2_D000X">
    <connection-url>jdbc:h2:mem:TGKDUMMYDS;DB_CLOSE_DELAY=-1</connection-url>
    <driver>h2</driver>
    <security>
        <user-name>sa</user-name>
        <password>sa</password>
    </security>
</datasource>
```

Attention: if you are using a WIldfly/JBoss JakartaEE version the security tag is inline:

```
<security user-name="tagetikcpm" password="tagetikcpm"/>
```

And add a driver reference for H2 inside the <drivers> tag as follows:

```
<driver name="h2" module="com.h2database.h2">
    <xa-datasource-class>org.h2.jdbcx.JdbcDataSource</xa-datasource-class>
</driver>
```

Step 3 – System properties configuration.

System properties can be configured modifying the standalone.conf.bat (Windows) or the standalone.conf.sh (Linux) files.

JVM Configuration

The following example shows how to configure JVM settings:

Microsoft Windows: standalone.conf.bat

```
set JAVA_OPTS=-Xrs -server –Xmx12g -XX:+UseCodeCacheFlushing -XX:-TieredCompilation -
XX:ReservedCodeCacheSize=512m
```

Linux: standalone.conf

```
JAVA_OPTS="$JAVA_OPTS -Xrs -server -Xmx8192m -XX:+UseCodeCacheFlushing -XX:-TieredCompilation
-XX:ReservedCodeCacheSize=512m"
```

Note: The above JVM configuration should be used as an example. Please refer to the "Technical Architecture: CCH Tagetik 5.3 Hardware & Software Architecture Overview" document to perform accurate dimensioning.

Logging configuration

Application logging is implemented with log4j2 and can be configured using the standard log4j2 configuration setting.
Simply add the following system property to the standalone.conf.bat (Windows) or standalone.conf.sh (Linux) files:

Microsoft Windows: standalone.conf.bat

```
set JAVA_OPTS="$JAVA_OPTS -Dlog4j.configurationFile=file:///{WILDFLY_HOME}/log4j2.xml"
```

Linux: standalone.conf

```
JAVA_OPTS="$JAVA_OPTS -Dlog4j.configurationFile={WILDFLY_HOME}\log4j2.xml"
```

Note: Substitute the variable WILDFLY_HOME with the location where WildFly is installed.

The use of the legacy variable -Dit.grupposervizi.easy.log4j.configuration is supported up to SP31. Its functionality will not be guaranteed in subsequent releases. If your system still use this legacy system property please evaluate to use standard log4j2 config file property to avoid future issues.

Example of *log4j2.xml* file that prints to the console and to a text file using a rolling policy:

```
<?xml version="1.0" encoding="UTF-8"?>
<Configuration status="warn" monitorInterval="30">
  <Appenders>
    <Console name="ConsoleAppender" target="SYSTEM_OUT">
```

```
      <PatternLayout pattern="%-5r %-5p [%c] (%M:%L) %m%n"/>
    </Console>
    <RollingFile name="RollingFileAppender" fileName="/{WILDFLY_HOME}/standalone/log/tagetik.log"
filePattern="/{WILDFLY_HOME}/standalone/log/tagetik.log.%i" ignoreExceptions="false">
      <PatternLayout>
        <Pattern>%-5r %-5p [%c] (%M:%L) %m%n</Pattern>
      </PatternLayout>
      <Policies>
        <SizeBasedTriggeringPolicy size="100MB"/>
      </Policies>
      <DefaultRolloverStrategy max="3"/>
    </RollingFile>
  </Appenders>
  <Loggers>
    <Root level="info" name="root">
      <AppenderRef ref="ConsoleAppender"/>
      <AppenderRef ref="RollingFileAppender"/>
    </Root>
  </Loggers>
</Configuration>
```

Note: Log4j2 configuration templates can be found on the CCH Tagetik Community web site.

Import / Export configuration

Please add the following system property for configuring the directory to be used to store CCH Tagetik export/import files:

Microsoft Windows: standalone.conf.bat

```
set JAVA_OPTS="$JAVA_OPTS -Dit.grupposervizi.easy.impexp.writableFolder=file:///{WILDFLY_HOME}/tagetik/imp-exp"
```

Linux: standalone.conf

```
JAVA_OPTS="$JAVA_OPTS -Dit.grupposervizi.easy.impexp.writableFolder={WILDFLY_HOME}\tagetik\imp-exp"
```

Deployment Timeout Configuration

Application deployment could take longer than the default timeout. To increase the blocking timeout please add the `jboss.as.management.blocking.timeout` system property using an appropriate value in seconds.

Microsoft Windows: standalone.conf.bat

```
set JAVA_OPTS="$JAVA_OPTS -Djboss.as.management.blocking.timeout=900"
```

Linux: standalone.conf

```
JAVA_OPTS="$JAVA_OPTS -Djboss.as.management.blocking.timeout=900"
```

File Upload Configuration

By default, WildFly and JBoss EAP do not allow to upload files greater than 10MB. It is possible to increase this threshold by changing the value of the *max-post-size* attribute as explained at https://docs.wildfly.org/26.1/Admin_Guide.html#server-configuration.

Deploying CCH Tagetik Application

We can deploy applications in WildFly or JBoss EAP in several ways:

1. File system copy (standalone mode only)
2. Using the Management Interfaces (Management Console or CLI)

Please refer to the official AS documentation in order to deploy the CCH Tagetik application EAR file (Application Deployment documentation for WildFly can be found at https://docs.wildfly.org/26.1/Admin_Guide.html#application-deployment, please check the documentation version in the URL).

Glossary

The following table provides a short glossary for the terminology used in this manual.

AS	Application Server
DB	Database
RDBMS	Relational Database Management System, i.e., MS SQL Server, Oracle, PostgreSQL
JEE	Java/Jakarta Platform Enterprise Edition
EAR	JEE Enterprise Archive, a file format used in JEE enterprise applications.

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